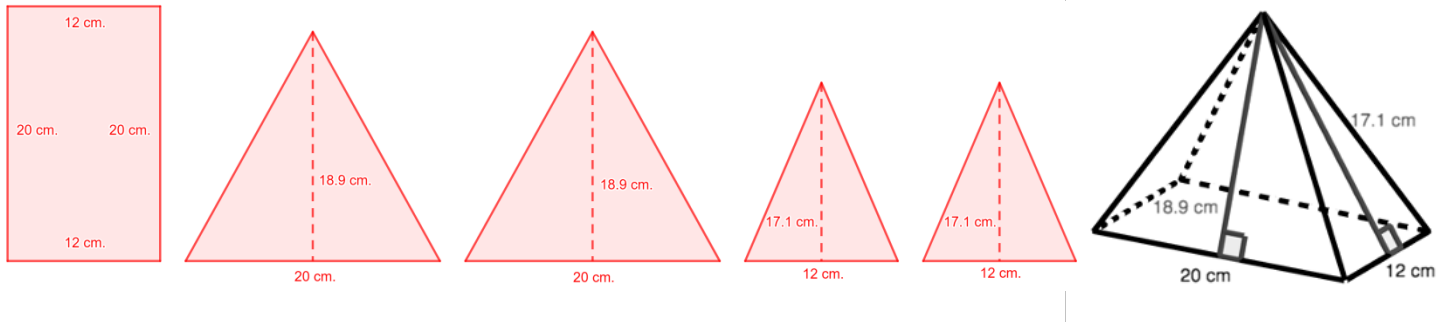


Grade 6
Unit 8
Lesson 12 Practice

Name _____
Date _____
Period _____

A rectangular pyramid is shown with measurements in centimeters (cm).

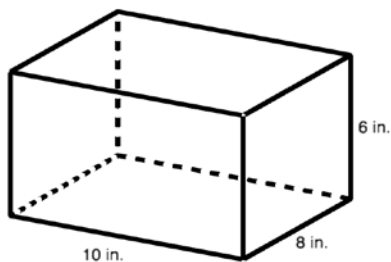
1. Draw a net or the individual faces of the figure. Label the measurements.



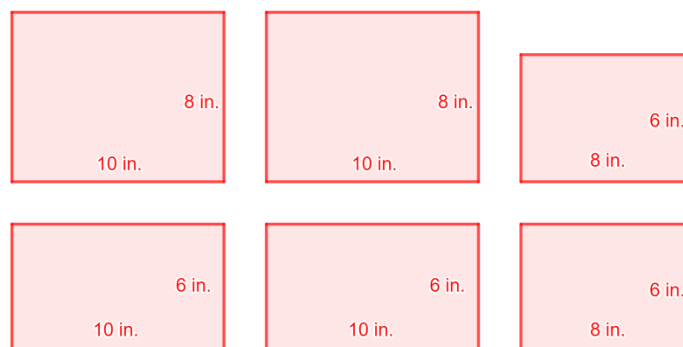
2. Find the surface area of the figure.

$$\begin{aligned}
 1. \quad 12 \times 20 &= 240 \text{ cm.}^2 & 4. \quad \frac{1}{2}(12)(17.1) &= 102.6 \text{ cm.}^2 \\
 2. \quad \frac{1}{2}(20)(18.9) &= 189 \text{ cm.}^2 & 5. \quad \frac{1}{2}(12)(17.1) &= 102.6 \text{ cm.}^2 \\
 3. \quad \frac{1}{2}(20)(18.9) &= 189 \text{ cm.}^2 & 240 + 2(189) + 2(102.6) &= 823.2 \text{ cm.}^2
 \end{aligned}$$

A rectangular prism is shown with measurements in inches (in.).



3. Draw a net or the individual faces of the figure. Label the measurements.



4. Find the surface area of the figure.

$$\begin{aligned}
 &(10 \times 8) + (10 \times 8) + (8 \times 6) + (8 \times 6) + (10 \times 6) + (10 \times 6) \\
 &= 80 + 80 + 48 + 48 + 60 + 60 = 376 \text{ in.}^2
 \end{aligned}$$

5. Find the volume of the figure.

$$\begin{aligned}
 V &= \ell \times w \times h \\
 V &= 10 \times 8 \times 6 \\
 V &= 480 \text{ in.}^3
 \end{aligned}$$

Tony designs earrings and is making post earrings in the shape of rectangular pyramids. They have a square base of 4 millimeters (mm). The slant height is 3.6 mm.

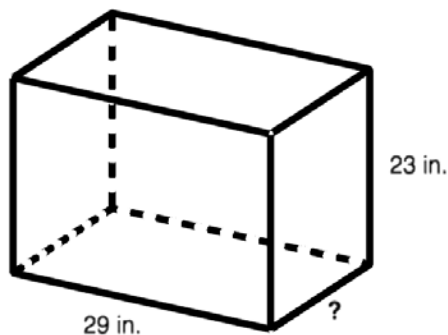
6. What is the surface area of one earring?

$$\begin{aligned} & (4 \times 4) + \left(\frac{1}{2} \times 4 \times 3.6\right) + \left(\frac{1}{2} \times 4 \times 3.6\right) + \left(\frac{1}{2} \times 4 \times 3.6\right) + \left(\frac{1}{2} \times 4 \times 3.6\right) \\ &= 16 + 7.2 + 7.2 + 7.2 + 7.2 \\ &= 44.8 \text{ mm.}^2 \end{aligned}$$

7. What is the surface area of a pair of earrings?

$$2(44.8) = 89.6 \text{ mm.}^2$$

Delilah is designing a toy box for her cousin. She drew a model of the toybox.



8. If the case has a volume of 12,673 cubic inches (in^3), what is the width of the box?

$$\begin{aligned} V &= \ell \times w \times h & 12,673 &= 667w \\ V &= 24 \times w \times 23 & \Rightarrow w &= 19 \text{ in.} \\ V &= 667w \end{aligned}$$

9. How much wood would Delilah need to make a box with these dimensions, not taking into consideration any overlap of the faces?

$$\begin{aligned} & (29 \times 19) + (29 \times 19) + (19 \times 23) + (19 \times 23) + (29 \times 3) + (29 \times 3) \\ &= 551 + 551 + 437 + 437 + 667 + 667 \\ &= 3310 \text{ in.}^2 \end{aligned}$$

10. If the inside of the box has a length of 28 in., a width of 16.5 in., and a height of 17 in., how much smaller is the volume of the inside of the toybox than the volume of the entire toybox?

$$\begin{aligned} V_{\text{inside}} &= 28 \times 16.5 \times 17 = 7,854 \text{ in.}^3 \\ 12,673 - 7,854 &= 4,819 \text{ in.}^3 \end{aligned}$$